

REFERENCE STUDY FOR VITAMATE

An Integrated Health and Lifestyle Tracking System with AI-Assisted Food Analysis

English Translation and Revised Academic Edition

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1. Introduction to the Reference Study

Digital health has developed rapidly in recent years as a result of advances in artificial intelligence, data analytics, wearable devices, mobile platforms, and intelligent applications. This evolution has produced systems that help individuals monitor multiple dimensions of daily health, including nutrition, physical activity, sleep, hydration, medication use, and lifestyle-related habits. Contemporary applications increasingly extend beyond simple data logging by providing structured feedback, reminders, progress visualization, and easier ways to capture health information.

Effective self-management of a healthy lifestyle depends on the user's awareness of daily behavior and the ability to observe and improve that behavior over time. Healthy living is not determined by a single factor; it is influenced by several interrelated dimensions, including:

- Food quality and the nutritional composition of consumed meals.
- Level of physical activity.
- Amount of water and other fluids consumed.
- Sleep quality, duration, and regularity.
- Medication adherence.
- Reduction of unhealthy habits such as smoking, excessive caffeine intake, excessive sugar intake, and frequent fast-food consumption.

Intelligent applications can support these areas by organizing daily health data, reducing the burden of manual tracking, and connecting related indicators. Artificial intelligence can add further value when it is used for well-scoped tasks such as food-image analysis, pattern extraction, and data-entry assistance. In VitaMate, AI is treated as an assistive component rather than an autonomous medical decision-maker.

VitaMate is designed as an integrated platform for health and lifestyle self-management. Rather than focusing on a single tracker, the project brings together several related services:

- Nutrition and detailed nutrient tracking.
- Daily hydration tracking.
- Physical-activity and step tracking.
- Sleep tracking.
- Medication tracking and reminder support.
- Tracking of unhealthy habits and personal goals.
- Consideration of selected chronic-condition constraints related to lifestyle and nutrition, including diabetes, hypertension, and lipid/cholesterol disorders.
- A motivation system based on verified actions, points, missions, streaks, achievements, and progress feedback.
- A coordinated notification system.
- AI-assisted food-image analysis that produces provisional nutritional estimates for user review before final logging.

The principal value of VitaMate lies in integrating these functions within one platform and coordinating information that would otherwise remain fragmented across separate applications. The project therefore emphasizes cross-domain organization, user-centered design, motivation, reminders, constraints-aware tracking, and an AI-assisted food-analysis workflow instead of an AI-generated diet-recommendation service.

This reference study reviews applications, research studies, datasets, and technical approaches that correspond to the major VitaMate domains. It aims to identify strengths and limitations in existing solutions and to clarify where VitaMate can add value through integration, chronic-condition awareness, motivational design, coordinated notifications, and AI-assisted meal analysis.

2. Study Problem and Reference Gap

Despite the widespread adoption of digital-health applications, many products remain specialized around one area such as food tracking, physical activity, sleep, hydration, medication adherence, or habit change. A user who wants to manage several health dimensions may therefore need multiple applications, resulting in fragmented records and limited cross-domain interpretation.

A second limitation is that personalization and coordination are often implemented only within a single feature. For example, an activity application may adapt activity goals without considering medication schedules, hydration, nutrition, or chronic-condition restrictions. Similarly, a nutrition application may provide detailed nutrient analysis while remaining disconnected from sleep, activity, and medication information.

The use of AI is also frequently restricted to individual functions, such as recognizing food from images. This can be valuable, but AI output should remain transparent, reviewable, and clearly separated from validated health rules. VitaMate therefore uses AI primarily to assist food logging, while health constraints and cross-domain coordination remain explicit system logic rather than unverified AI-generated medical advice.

The reference gap addressed by VitaMate can therefore be summarized as the limited availability of a single platform that combines nutrition, hydration, activity, sleep, medication, unhealthy-habit tracking, chronic-condition constraints, motivation, coordinated notifications, and review-assisted food-image analysis within one consistent software architecture.

3. Objectives of the Reference Study

- Analyze health and lifestyle applications that are relevant to the VitaMate scope.
- Examine how major VitaMate services are implemented in existing applications.
- Identify publicly described technologies, algorithms, and design approaches.
- Summarize the strengths and limitations of existing solutions.
- Assess the role of artificial intelligence in food-image analysis and digital dietary assessment.
- Add a research-based reference for motivation and gamification mechanisms relevant to long-term engagement.
- Identify market and research gaps that remain insufficiently integrated.
- Use the findings to support VitaMate design decisions on an academic and evidence-aware basis.

4. Methodology of the Reference Study

The study follows a descriptive and analytical methodology. Information was collected from scientific literature, peer-reviewed publications, official application websites, application-store descriptions, and publicly available technical documentation. The review covers the domains represented in VitaMate, including nutrition, activity, hydration, sleep, medications, unhealthy-habit support, motivation, and AI-assisted food analysis.

Each reference application was analyzed using a consistent set of criteria: the main services provided, publicly described technologies or mechanisms, apparent methodological approach, major strengths, major limitations, and relevance to VitaMate. A comparative matrix was then used to identify which health domains are strongly, partially, or weakly represented across the reviewed applications.

The motivation section was strengthened with systematic reviews and recent evidence on gamification in mobile health, in addition to Duolingo as a widely recognized non-health benchmark for habit formation, streaks, quests, points, feedback, social motivation, and notification timing. Duolingo is included as a behavioral-design reference rather than as a health application.

The AI food-analysis section preserves the research material from the original study and organizes it around datasets, computer-vision models, portion/weight estimation, nutritional estimation, and comparative performance. The previously included AI-generated smart-diet/recommendation section has been removed because it is no longer part of the implemented VitaMate scope.

5. Reference Applications

A key challenge in the VitaMate reference study is that the project spans functions that are typically distributed across separate applications. Consequently, the benchmark is intentionally multi-source: nutrition applications are used to study nutrient tracking and food logging; medication applications are used for adherence and reminder logic; hydration and sleep applications provide specialized interaction patterns; activity applications provide progress and engagement mechanisms; and motivation-oriented products provide reusable design ideas for sustained participation.



MyNetDiary

MyNetDiary / PlateAI

What it provides

- Calorie and macronutrient tracking.
- Tracking of more than 100 micronutrients and detailed nutritional elements.
- Personalized meal-plan features.
- Food logging by image and voice.
- AI Coach functionality.
- Menu Scan for restaurant-related suggestions.
- Health tracking related to diabetes, including blood glucose, insulin, blood pressure, and cholesterol within its broader health ecosystem.

Technologies / mechanisms

- AI Meal Scan for image-based food analysis.
- Voice Logging for spoken food entry.
- AI Coach for personalized support.
- Menu Scan for restaurant-menu analysis.
- A large verified food database.

Publicly described methodology

- The application publicly presents several AI-powered capabilities, but does not fully disclose specific model architectures.
- Its visible approach combines image analysis, nutrition databases, coaching/support features, and selected chronic-condition indicators.

Strengths

- One of the closest references to the broad VitaMate nutrition vision because it goes beyond calories to micronutrients and selected chronic-health indicators.

Limitations

- It remains primarily a general nutrition platform; the public material does not demonstrate a centralized, cross-tracker constraints layer equivalent to VitaMate.
- Detailed algorithmic information is not publicly disclosed.

Relevance to VitaMate

- Useful as a reference for combining detailed nutrition tracking, AI-assisted logging, and selected chronic-condition information.
 - VitaMate differentiates itself through cross-domain integration, centralized constraints, motivation, and coordinated notification behavior rather than AI-generated diet prescriptions.
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Cronometer

What it provides

- Calorie and macronutrient tracking.
- Extensive micronutrient tracking.
- Dietary targets based on Dietary Reference Intakes (DRIs).
- Manual customization of nutritional targets.
- Minimum and maximum thresholds for individual nutrients.
- Detailed reports comparing intake with daily requirements.

Technologies / mechanisms

- DRI-based nutrient targeting.
- Custom targets for vitamins, minerals, and macronutrients.
- Detailed nutrition reports.
- Threshold tracking for minimum and maximum nutrient values.

Publicly described methodology

- The core methodology relies on nutritional reference standards, customizable targets, analytics, and reports rather than AI image recognition.

Strengths

- A strong benchmark for nutrition systems that go beyond calorie counting.
- Particularly useful for vitamins, minerals, electrolytes, and daily nutrient thresholds.

Limitations

- AI-assisted photo logging is not the primary focus.
- Availability of micronutrient data can vary according to the underlying food source.

Relevance to VitaMate

- Provides a strong reference for complete nutrient-requirement logic and comparison of actual intake against detailed nutrient needs.
- VitaMate adds AI-assisted food entry and cross-domain health context around this type of nutrient tracking.



Nutrivine

What it provides

- Tracking of more than 60 nutritional elements.
- A food database advertised at more than four million foods.
- Personalized DRI-based goals using age, sex, and life stage.
- AI food scanning.
- Barcode scanning.
- Voice logging.
- AI insights for identifying dietary patterns and nutrient gaps.

Technologies / mechanisms

- AI Food Scanning for foods and products.
- Barcode scanning for packaged products.
- Voice logging.
- DRI-based personalization.
- AI insights for pattern and nutritional-gap analysis.

Publicly described methodology

- The product is publicly described as AI-powered, although specific architectures such as CNNs or Transformers are not disclosed.
- Its stated approach combines AI scanning, personalized DRI targets, and nutritional-gap analysis.

Strengths

- Closely aligned with VitaMate in combining micronutrients with AI-assisted food entry.
- Links tracking to reference-based nutritional goals rather than calories alone.

Limitations

- Public information is less explicit regarding chronic-condition coordination.
- It does not clearly describe how disease-related constraints are translated into cross-module system rules.

Relevance to VitaMate

- Useful for combining micronutrient tracking with AI-assisted input.
- VitaMate extends the concept through centralized constraints, cross-tracker coordination, and a broader health-management scope.



Microgram

What it provides

- Food analysis from image or voice input.
- Tracking of more than 50 nutritional elements.
- Tracking of vitamins, minerals, amino acids, and omega-related nutrients.
- A daily personalized score.
- Explanations of how individual nutrients affect the body.
- Customization of visible nutrients according to user interest.

Technologies / mechanisms

- Computer Vision for food recognition.
- Natural Language Processing for user input.
- Depth Perception for portion-size estimation.
- Voice AI for spoken logging.
- USDA/NCCDB matching for nutrition lookup.

Publicly described methodology

- Public descriptions reference computer vision, NLP, and depth-perception techniques without naming a specific model.
- The apparent pipeline is image capture → portion estimation → database matching → nutrient calculation.

Strengths

- A strong reference for combining AI food imaging with detailed micronutrient tracking.
- Directly relevant to automated estimation of calories and nutritional values from food images.

Limitations

- Chronic-condition support is less explicit.
- The product is closer to intelligent nutritional tracking than to a fully integrated health platform.

Relevance to VitaMate

- A particularly useful reference for the AI-assisted meal-image feature.
- VitaMate combines this type of analysis with user review, nutrition logging, constraints-aware tracking, and other health domains.



Foodvisor

What it provides

- Manual food logging.
- Photo-based meal analysis in paid tiers.
- Food diary.
- Customizable macronutrient goals.
- Daily summaries.

- Articles, meal plans, and recipes.
- Daily lessons related to nutrition and motivation.

Technologies / mechanisms

- Photo meal analysis.
- Manual logging.
- Barcode scanning.
- Goal customization.
- Daily summary and coaching-style feedback.

Publicly described methodology

- The product publicly describes image analysis but does not expose detailed AI model architectures.
- The visible experience combines photo-based logging, daily coaching, and meal-planning content.

Strengths

- Strong user experience.
- Combines tracking, image-based logging, and daily educational/motivational content in a practical workflow.

Limitations

- Less detailed than Cronometer or MyNetDiary in micronutrient depth.
- Chronic-condition coordination is not a central publicly described capability.

Relevance to VitaMate

- Useful as a UI/UX reference for simplifying food entry and presenting daily feedback.
 - VitaMate extends the experience with broader health tracking, micronutrients, constraints-aware coordination, and a distinct motivation system.
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Medisafe

What it provides

- Medication-time reminders.
- Refill reminders.
- Drug-interaction warnings.
- Medication tracking for family members.
- Reports that can be shared with clinicians.
- Additional health measurements alongside medication tracking.

Technologies / mechanisms

- Medication scheduling by time and treatment schedule.
- Refill reminders.
- Drug-interaction warning mechanisms.
- Family tracking for more than one person.

Publicly described methodology

- Publicly presented as a medication-adherence and scheduling platform with reminder, safety, and reporting features.

Strengths

- Very strong medication-management reference, especially for complex schedules, reminders, and adherence support.

Limitations

- Medication management is the main focus; deep coordination with nutrition, hydration, sleep, and lifestyle habits is limited compared with VitaMate's integrated scope.

Relevance to VitaMate

- Useful for medication reminder logic, dose tracking, and reporting patterns.
- VitaMate places medication records inside a broader health platform without attempting to replace clinical medication decisions.



MyTherapy

MyTherapy

What it provides

- Medication reminders.
- Automatic documentation of taken or missed doses.
- Personal notes for individual doses.
- Symptom and measurement tracking.
- Refill reminders.
- Report-based sharing of treatment records with clinicians.

Technologies / mechanisms

- Medication reminders.
- Health diary for doses, symptoms, and measurements.
- Dose documentation.
- Report sharing.

Publicly described methodology

- Combines adherence reminders with a longitudinal health diary and report-oriented tracking.

Strengths

- Links medication adherence and daily health journaling more clearly than many medication-only applications.

Limitations

- Still focused primarily on treatment adherence rather than a full platform linking medication with nutrition, activity, hydration, sleep, and motivation.

Relevance to VitaMate

- A useful model for combining medication records with health journaling.
- VitaMate broadens this idea through cross-domain coordination and a unified notification infrastructure.



WaterMinder

What it provides

- Water and fluid tracking.
- Personalized daily hydration goals.
- Flexible reminders based on the daily schedule.
- Daily, weekly, and monthly history.
- Caffeine tracking.
- Support for Apple Watch, Apple Health, and Google Fit.

Technologies / mechanisms

- Personalized hydration goals.
- Smart reminders.
- Drink logging across multiple beverage types.
- History and trend visualization.

Publicly described methodology

- A focused hydration workflow built around personalized goals, reminders, fast logging, and long-term progress review.

Strengths

- A clear and widely recognized reference for hydration tracking, ease of use, and personalization.

Limitations

- Focuses mainly on hydration and does not deeply coordinate water intake with chronic conditions or the complete health context.

Relevance to VitaMate

- Useful for hydration logging, progress visualization, and reminder design.
- VitaMate integrates hydration with activity, medication, nutrition, caffeine, motivation, and shared health-state information.



Hydro Coach

What it provides

- Water and fluid tracking.

- Hydration reminders.
- Support for multiple beverage types.
- Integration with Apple Health, Fitbit, Samsung Health, and Google Fit.
- CSV data export.
- Apple Watch support.

Technologies / mechanisms

- Smart Drink Reminder.
- Interval-based reminders.
- Drink-type logging.
- Health-platform synchronization.

Publicly described methodology

- A hydration-centric model that emphasizes reminders and interoperability with external health platforms.

Strengths

- A useful reference for integrating hydration data with other health ecosystems.

Limitations

- Like many hydration applications, it does not build a broader cross-domain health model influenced by sleep, medication, activity, and nutrition.

Relevance to VitaMate

- Useful for platform synchronization concepts.
- VitaMate treats hydration as one component of a wider health-management system.



Sleep Cycle

Sleep Cycle

What it provides

- Sleep tracking.
- Smart Alarm designed to wake the user during a lighter sleep phase.
- Sleep-stage analysis.
- Recording of sleep sounds such as snoring or talking.
- Sleep Aid audio and stories.
- Integration with iOS 18 and watchOS 11 as described in the original study.

Technologies / mechanisms

- Smart Alarm within a defined wake-up window.
- Sleep-stage analysis for light, deep, and REM sleep.
- Sound recording.
- Sleep-aid audio.

Publicly described methodology

- A sleep-centered analysis workflow that combines sleep-session monitoring, stage interpretation, acoustic signals, and smart alarms.

Strengths

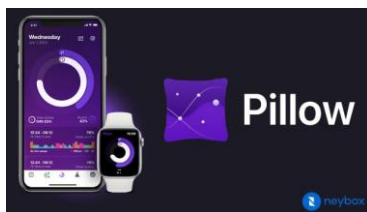
- Strong sleep-analysis and wake-up experience.

Limitations

- Sleep is treated primarily as a standalone service rather than as one element in a multi-domain health platform.

Relevance to VitaMate

- Useful for sleep-session design, alarms, and quality summaries.
- VitaMate uses sleep records as part of broader progress and health-state coordination rather than as an isolated feature.



Pillow

What it provides

- Automatic or manual sleep tracking.
- Sleep score.
- Analysis of time in bed versus actual sleep time.
- Heart-rate tracking.
- Apple Health integration.
- Support for Siri, Shortcuts, Widgets, and Live Activities.
- Apple Watch support.

Technologies / mechanisms

- Automatic sleep tracking.
- Manual sleep sessions when needed.
- Sleep score.
- Health integration with Apple Health and watch data.

Publicly described methodology

- A flexible sleep-tracking model centered on automatic capture, optional manual correction, quality scoring, and ecosystem integration.

Strengths

- Highly flexible automatic tracking and strong integration within the Apple ecosystem.

Limitations

- Primarily a sleep service and not an integrated chronic-condition or multi-domain health-management platform.

Relevance to VitaMate

- Useful for sleep logging, quality scoring, and integration patterns.
 - VitaMate combines sleep with other trackers and uses it in cross-domain progress summaries.
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Strava

What it provides

- Recording of sports activities.
- Tracking of running, cycling, walking, and many other activities.
- Support for more than 50 activity types as reported in the original study.
- Challenges and a sports community.
- Recording through phones or wearable devices.
- Progress and achievement tracking.

Technologies / mechanisms

- Activity recording for route, duration, and performance.
- Multi-sport support.
- Challenges to encourage adherence.
- Community and social tracking.

Publicly described methodology

- Combines activity measurement with achievements, challenges, and social motivation.

Strengths

- A strong reference for activity adherence, community participation, challenges, and motivational feedback.

Limitations

- Focused on physical activity rather than deep integration with nutrition, medication, sleep, chronic-condition constraints, and other health data.

Relevance to VitaMate

- Useful for activity tracking and motivational design.
 - VitaMate adopts motivation around verified health actions while keeping the system non-competitive where competition could distort healthy behavior.
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Google Fit

What it provides

- Daily activity tracking.
- Step tracking.
- Heart Points in collaboration with WHO and AHA, as described in the original reference study.
- Walking, running, and cycling tracking.
- Speed, pace, and route display.
- Integration with multiple health applications and devices, including services such as Strava, MyFitnessPal, and Lifesum.

Technologies / mechanisms

- Heart Points system for activity goals.
- Workout journaling.
- Cross-application integration.
- Daily step and activity-goal tracking.

Publicly described methodology

- A general activity-aggregation model that emphasizes unified activity records and connections with other platforms.

Strengths

- A useful reference for aggregating activity information from multiple sources into one view.

Limitations

- General-purpose activity aggregation does not by itself provide deep coordination with nutrition, medications, chronic-condition constraints, or a project-specific motivation layer.

Relevance to VitaMate

- Useful as a reference for data aggregation and activity-goal concepts.
- VitaMate applies a similar integration principle across a broader set of health domains.

5.1 Cross-Application Benchmark Observations

The application review shows a recurring specialization pattern. Nutrition products provide the richest nutrient databases and food-entry options; medication products concentrate on adherence and scheduling; hydration and sleep products optimize narrow daily routines; and activity platforms combine tracking with goals, challenges, and social feedback. This specialization explains why no single benchmark should be treated as a complete substitute for VitaMate.

A second pattern is that strong user experience often comes from reducing the effort required to log an action. Image or voice food entry, one-tap medication confirmation, reminder-driven hydration logging, automatic sleep capture, and passive activity recording all reduce friction. For VitaMate, this supports a

design strategy in which frequently repeated actions are kept short, visually clear, and connected to immediate progress feedback.

A third pattern concerns motivation. Activity and habit-oriented products are more likely to use streaks, achievements, challenges, progress indicators, or social reinforcement, while nutrition and medication products are generally more conservative. The expanded motivation section therefore complements the health-app comparison with three dedicated behavioral-design benchmarks—Duolingo, Habitica, and Finch—before the AI food-analysis review.

5.2 Cross-Application Feature Matrix

The matrix below preserves the original benchmark coverage and adds a motivation/gamification column to reflect the updated VitaMate scope. Duolingo is included only as a motivation-design benchmark; it is not classified as a health application.

Application	Nutrition	Hydration	Activity	Sleep	Medication	Unhealthy habits	Chronic conditions	AI	Motivation / Gamification
MyNetDiary / PlateAI	✓	X	X	X	X	X	△	✓	△
Cronometer	✓	X	X	X	X	X	X	X	X
Nutrivine	✓	X	X	X	X	X	△	✓	△
Microgram	✓	X	X	X	X	X	X	✓	△
Foodvisor	△	X	X	X	X	X	X	✓	△
Medisafe	X	X	X	X	✓	X	X	X	△
MyTherapy	X	X	X	X	✓	X	X	X	△
WaterMinder	X	✓	X	X	X	△	X	X	△
Hydro Coach	X	✓	X	X	X	X	X	X	△
Sleep Cycle	X	X	X	✓	X	X	X	X	△
Pillow	X	X	X	✓	X	X	X	X	△
Strava	X	X	✓	X	X	X	X	X	✓
Google Fit	X	X	✓	X	X	X	X	X	△
BeSober	X	X	X	X	X	✓	X	X	✓
Quitzilla	X	X	X	X	X	✓	X	X	✓
We Quit!	X	X	X	X	X	✓	X	X	✓
Addicted: Quit Habit Tracker	X	X	X	X	X	✓	X	X	✓
QuitWithUs	X	X	X	X	X	✓	X	X	✓
Duolingo	X	X	X	X	X	X	X	X	✓
Habitica	X	X	X	X	X	✓	X	X	✓
Finch	△	△	△	△	X	△	X	X	✓

Legend: ✓ clearly present △ partially present / indirect X not a core feature

The matrix shows that most benchmark products remain domain-focused. The added motivation benchmarks occupy a different role: Duolingo demonstrates long-term streak, quest, and progress mechanics; Habitica demonstrates explicit points, levels, rewards, and role-playing progression; and Finch demonstrates a gentler self-care model based on a companion, mini-goals, quests, milestones, and flexible streaks. These references are therefore used to study engagement design rather than to classify them as direct health-platform competitors.

6. Motivation and Gamification Reference

Motivation became a major development focus in the later VitaMate stage, so the reference study was expanded to examine evidence on gamification and behavior-change design. Gamification refers to the use of game-design elements in non-game contexts to support engagement, adherence, and repeated behavior. Common mechanisms include goal setting, points, progress bars, streaks, badges, levels, challenges, quests, feedback, reminders, and social support.

In health applications, these elements should not be treated as decorative rewards. Their value depends on whether they reinforce the target health behavior, provide understandable feedback, support self-monitoring, and avoid encouraging unhealthy competition or reward-seeking that is disconnected from the health objective.

6.1 Evidence from Health and Behavior-Change Research

Study	Type / Scope	Main Finding	Relevance to VitaMate
Edwards et al., 2016	Systematic review of gamification in health-promotion smartphone apps	Among 64 gamified health apps, the dominant behavior-change categories were feedback/monitoring, rewards, and goals/planning. The study emphasized wide variation in implementation and limited evidence linking more game elements to better health outcomes.	Supports combining self-monitoring, feedback, goals, and rewards rather than relying on points alone.
Sardi et al., 2017	Systematic review of gamification in e-health	Reviewed 46 studies across health and wellness contexts. Reported potential motivational benefits but noted limited empirical evidence and the risk that extrinsic rewards may mainly generate short-term engagement.	Supports a theory-aware motivation system and cautions against purely extrinsic scoring.
Xu et al., 2022	Systematic review of mHealth gamification for physical activity	Reviewed 50 studies. Common elements included goal setting, progress bars, rewards, points, and feedback. Effects on physical activity were generally positive but mixed and often modest.	Supports goals, visible progress, rewards, and feedback while keeping expectations realistic.
Tran et al., 2022	Scoping review of gamification/incentives for medication adherence	Included 11 studies; gamification was used in most. Outcomes were generally improved or maintained, but study designs and populations were heterogeneous. Users also raised concerns about repetitive or irrelevant features.	Relevant to medication adherence, reminders, and the need for meaningful, user-centered motivation.
Nishi et al., 2024	Systematic review and meta-analysis of randomized trials of health apps with/without gamification	Included 36 trials and more than ten thousand participants. The evidence indicates that effects vary by outcome, population, and intervention design rather than gamification acting as a universal solution.	Supports measured use of gamification as one component of a broader behavior-change system.
Gerdenitsch et al., 2020	Survey of 114 employees using Habitica for work-related tasks	Participants used Habitica for organization, self-monitoring, and self-motivation. The study reported positive effects on work enjoyment and context-dependent productivity effects.	Shows how a mature gamified task system can support organization and repeated behavior, while effects depend on user context.
Diefenbach and Müssig, 2019	Qualitative and field studies of Habitica users	The authors identified counterproductive gamification effects when reward or punishment mechanics conflicted with the intended behavior or users adapted tasks to game the system.	Reinforces the need for VitaMate rewards to follow verified healthy actions and to avoid punitive or exploitable scoring.

6.2 Duolingo as a Motivation-Design Benchmark

duolingo

Duolingo

Behavioral-design benchmark (not a health application)

Duolingo is included because it provides a mature example of how an application can transform a long-term, effortful behavior into a repeatable daily routine. Its official product material describes a combination of experience points (XP), streaks, daily and monthly quests, leaderboards/leagues, friend quests, milestone celebrations, and personalized reminders. Duolingo also publicly reports extensive A/B testing of

engagement mechanics and has modified features when reward optimization risked distracting users from the desired learning behavior.

- XP and levels provide immediate feedback after completed actions.
- Streaks visualize continuity and reinforce daily habit formation.
- Daily and monthly quests convert broad goals into small, concrete tasks.
- Leaderboards create optional social comparison and friendly competition.
- Friends Quests and social features introduce accountability and cooperation.
- Milestones and celebrations make long-term progress visible.
- Personalized reminders support re-engagement when the user is likely to miss a routine.
- Streak Freeze introduces controlled flexibility so that one missed day does not necessarily destroy a long-term habit.

The Duolingo example is particularly useful because it also demonstrates the risks of gamification. Research on Duolingo has documented “gamification misuse,” in which users can become overly focused on points, leaderboards, or competitive behavior rather than the underlying learning objective. Duolingo itself has described product changes intended to reduce XP grinding and to emphasize meaningful learning activity. This reinforces an important VitaMate design principle: motivational rewards should support healthy behavior, not become a substitute for it.

6.3 Habitica as a Gamified Habit-Formation Benchmark

HABITICA

Habitica

Gamified habit and task-management benchmark

Habitica transforms real-world habits, recurring daily tasks, and to-do items into a role-playing progression system. Its official feature material describes rewards such as Gold, Experience, collectible items, character equipment, and progression tied directly to task completion [48]. Unlike a simple points counter, the design connects repeated actions to a persistent game state, giving users a visible sense of advancement over time.

- Habits, Dailies, and To-Dos separate recurring behavior from one-time tasks.
- Gold and Experience provide immediate feedback for completed actions.
- Levels, equipment, collectibles, and quests create longer-term progression.
- Parties and social features introduce cooperation and accountability.
- Reward and punishment mechanics make missed actions visible, but can also create pressure or unintended incentives.

Habitica is supported by empirical research that is especially useful for VitaMate design. Gerdenitsch et al. studied 114 employees who used Habitica to gamify work tasks and found that users commonly relied on it for organization, self-monitoring, and self-motivation; effects on productivity were context-dependent [49]. Diefenbach and Müssig also documented counterproductive effects in Habitica, showing that poorly aligned reward or punishment rules can encourage gaming the system or reduce motivation [50]. These findings support VitaMate’s decision to reward verified health actions, prevent duplicate scoring, and avoid punitive mechanics that could discourage users after missed goals.

6.4 Finch as a Gentle Self-Care Motivation Benchmark



Finch

Finch provides a contrasting motivation model built around gentle self-care rather than competition. Official Finch documentation describes small customizable goals, a virtual companion (“birb”) that grows with the user, energy and Rainbow Stone rewards, daily quests, weekly milestones, seasonal events, streaks, reminders, social encouragement, and flexible mechanisms such as streak repair and pause mode [51]–[53]. The design attempts to make consistency feel supportive instead of punitive.

- Small daily goals reduce the activation cost of beginning a self-care routine.
- Completing goals energizes the companion and provides immediate emotional and visual feedback.
- Daily Quests and weekly milestones convert broad self-care intentions into short, attainable challenges.
- Rainbow Stones, items, adventures, and seasonal rewards create light extrinsic reinforcement.
- Streak repair and Pause Mode reduce the “all-or-nothing” pressure that can make rigid streak systems discouraging.
- Friends and positive social interactions support encouragement without requiring competitive leaderboards.

Finch is relevant to VitaMate because health behavior often requires flexibility. A missed hydration, activity, or sleep target should not make a user feel that all previous progress has been lost. The Finch model therefore supports a more compassionate interpretation of streaks and reminders: continuity should be encouraged, but recovery from missed days should be possible. This principle is particularly appropriate in a health-oriented system where illness, travel, treatment changes, or personal circumstances may temporarily interrupt routines.

6.5 Comparative Motivation Benchmarks

Benchmark	Primary Mechanisms	Main Design Strength	Important Risk / Limitation	Most Relevant VitaMate Lesson
Duolingo	XP, streaks, quests, leagues, milestones, reminders	Highly visible progress and strong daily-return loops	Users may optimize points or competition rather than the underlying learning goal	Make progress visible, but ensure rewards remain subordinate to the health behavior.
Habitica	Habits/dailies, XP, gold, levels, gear, quests, social parties	Persistent game progression tied to real-world tasks	Punishment or exploitable point rules can produce counterproductive behavior	Reward only verified actions; use idempotency and avoid punitive health mechanics.
Finch	Mini-goals, companion growth, quests, stones, milestones, streak repair, reminders	Gentle, emotionally supportive consistency and recovery	Rich visual/game layers may be distracting for users who prefer minimal interfaces	Use supportive feedback, flexible streaks, and recovery after missed days.

Taken together, the three benchmarks suggest that motivation should combine immediate feedback, visible progress, meaningful milestones, and manageable challenges while preserving user autonomy. Duolingo demonstrates the power of repeated progress loops; Habitica demonstrates deep task-to-reward mapping and the risks of misaligned incentives; Finch demonstrates a softer model that supports recovery and emotional reinforcement. VitaMate can combine these lessons without copying any one system wholesale.

6.6 Design Implications for the VitaMate Motivation System

- Reward verified health-related actions rather than screen time or arbitrary interaction.
- Use points as motivational feedback, while keeping them separate from health scores and health-state indicators.
- Use daily missions to convert broad health goals into achievable tasks.
- Use streaks to reinforce consistency, with enough flexibility to avoid punitive loss after unavoidable interruptions.

- Use badges and levels as milestone recognition rather than as indicators of medical status.
- Use a Motivation Feed to explain why a reward or achievement was earned and to make progress understandable.
- Connect notifications to meaningful reminders and unfinished missions instead of sending excessive generic prompts.
- Avoid health-risky competition, especially when users differ in medical conditions, activity capacity, or treatment constraints.
- Monitor for reward loops that could incentivize excessive logging, duplicate actions, or behavior chosen only to gain points.

This evidence-based framing supports the current VitaMate motivation architecture, in which points, missions, streaks, badges, levels, and feedback operate as an engagement layer around verified tracker actions. The motivation system is therefore designed as a behavioral-support mechanism, not as a clinical decision system.

7. AI-Assisted Food-Image Analysis Reference

Food-image understanding has been studied for decades, but advances in computer vision, deep learning, depth estimation, and large food datasets have made automatic meal analysis increasingly practical. The core problem remains challenging: an image does not directly reveal exact ingredient identity, preparation method, hidden ingredients, or true portion weight. Therefore, the strongest systems combine visual recognition with segmentation, portion or mass estimation, and nutrition-database lookup, while retaining a mechanism for uncertainty handling and user review.

For VitaMate, this research is relevant to reducing the burden of manual meal entry. The implemented direction is review-assisted rather than fully autonomous: the AI service generates provisional food information, the user can verify or correct uncertain fields, and the final nutrition record is created through the application’s authoritative nutrition workflow.

7.1 Nutrition5k: Towards Automatic Nutritional Understanding of Generic Food

Nutrition5k is a dataset introduced by Google Research for training and evaluating computer-vision systems that estimate the nutritional composition of real-world dishes. The study addressed a central limitation in image-based dietary assessment: internet food images are abundant, but accurate nutritional ground truth is scarce. Nutrition5k therefore collected approximately 5,000 real dishes under a controlled acquisition protocol and paired visual data with ingredient-level weights and nutritional annotations.

Publication: Nutrition5k was released as an arXiv preprint in March 2021 and published at CVPR 2021 by Quin Thames and colleagues at Google Research.

Dataset Composition and Modalities

- Approximately 5,000 real-world dish samples.
- RGB images and, for many samples, depth information (RGB-D).
- Multiple views or turntable-style visual captures in parts of the dataset.
- Measured ingredient weights that provide ground-truth mass information.
- Ingredient lists and recipe-related metadata.
- Ground-truth calories and macronutrients derived from measured ingredient quantities and nutritional databases.

Collection and Annotation Protocol

- Ingredients were physically weighed during dish preparation, providing stronger supervision than image labels alone.
- Multiple viewpoints and depth measurements improve the ability to study portion and volume estimation.
- Annotations include ingredient identity, ingredient weight, and nutritional values at dish/component level depending on the data record.
- Subsequent work has noted that data cleaning and annotation review remain important because real datasets can contain noise or inconsistencies.

Supported Research Tasks

- Food classification and ingredient recognition.
- Segmentation or detection of food components.
- Volume and weight estimation using RGB, depth, or learned geometric cues.
- Nutrient estimation by mapping estimated components and quantities to nutrition databases.

Strengths and Limitations

- Strength: measured ingredient weights provide unusually valuable ground truth.
- Strength: depth data supports research on volume and mass estimation.
- Limitation: approximately 5,000 dishes remain small relative to global cuisine diversity.
- Limitation: dependence on depth at inference time can reduce practicality on devices without reliable depth sensors.
- Limitation: data cleaning and domain adaptation may be needed before training a production-oriented model.

Relevance to VitaMate

Nutrition5k provides a strong research basis for training and evaluating the food-analysis pipeline in VitaMate. A practical workflow can use visual recognition to identify candidate foods, estimate portion or mass, and then map confirmed food items to a trusted nutritional database. In VitaMate, the final step remains review-assisted: estimates are not treated as unquestionable nutrition facts, and the user can confirm or correct the provisional result before it is saved.

7.2 2D Nutritional Composition Predictor (2DNCP)

The 2D Nutritional Composition Predictor reference in the original study corresponds to recent work evaluating deep-learning architectures for predicting food mass and nutritional composition from two-dimensional meal images. The study uses Nutrition5k-derived targets and compares well-known visual backbones, including ResNet-50/101, InceptionV3, and ViT-B/16, with regression heads for nutritional prediction.

Experimental Design

- Dataset splitting into training, validation, and test sets with attention to food diversity.
- Image preprocessing such as resizing, normalization, and data augmentation.
- Linking dish images with measured mass and nutritional targets.
- Regression heads attached to image encoders to estimate mass, calories, and macronutrients.
- Comparison of convolutional networks with Vision Transformer architectures.
- Evaluation using error metrics such as MAPE, MAE, or RMSE depending on the target.

Main Findings

- ResNet-101 and ViT-B/16 were reported among the strongest architectures in the referenced comparison.
- Replacing weaker visual backbones with stronger representations improved nutrient-estimation error for major targets.
- Depth information can improve portion/mass estimation when available, but RGB-only use is operationally easier on standard phones.
- Complex mixed dishes remain difficult because visual appearance does not uniquely determine ingredients or quantities.

For VitaMate, the study is useful when selecting the visual backbone and deciding whether performance gains from depth-dependent methods justify additional capture requirements. The project can favor a robust RGB-first workflow while keeping depth-assisted estimation as an optional or future enhancement.

7.3 Food Recognition and Calorie Estimation System (FRCES)

The original reference study describes FRCES as a multi-stage computer-vision pipeline for analyzing meal images and estimating basic nutritional values. The conceptual pipeline combines a food-recognition network such as EfficientNet, an object-detection model such as YOLOv8, and nutritional lookup using databases such as USDA FoodData Central.

- Input: a meal image.
- Food-type recognition and candidate classification.
- Detection or separation of multiple food components.
- Portion-size estimation when sufficient visual information is available.
- Nutrition lookup to estimate calories, protein, carbohydrates, and fat.
- Potential use in diet-tracking, chronic-condition self-management support, and digital-health applications.

The main relevance to VitaMate is architectural: image recognition alone is not enough. A practical service requires multiple stages and a trusted nutritional mapping layer. The original study did not provide a single verified benchmark figure for the complete FRCES pipeline, so performance should be evaluated using the specific models and datasets used in the VitaMate implementation.

7.4 Im2Calories

Im2Calories, developed by Google Research in 2015, is one of the early influential systems for estimating food energy from images. It introduced an end-to-end workflow in which foods are recognized, segmented, assigned portion or volume estimates, and mapped to nutritional information.

Element	Description
Model name	Im2Calories
Developer	Google Research
Year	2015
Field	Food Image Analysis
Model type	Deep Learning + Computer Vision

Pipeline

1. Recognize food items in the meal image.
2. Segment food components inside the plate or scene.
3. Estimate food volume / portion size.
4. Map food identity and estimated quantity to a nutrition database.

5. Calculate calories and related nutritional values.

Im2Calories remains important as a foundational reference because it formalized the “image → food identity → portion → nutrition” pipeline that is still visible in modern systems. For VitaMate, the lesson is to treat food recognition and portion estimation as separate uncertainties and to allow user correction before final nutrition logging.

7.5 Recipe1M+

Recipe1M+ is a large-scale dataset created to connect food images with recipe text and ingredient information. It is especially valuable for cross-modal learning, ingredient recognition, and image-to-recipe retrieval rather than direct portion estimation.

Element	Description
Project name	Recipe1M+
Developed by	MIT / collaborating research teams
Publication year	2017
Field	Food Recognition and Cross-Modal Recipe Understanding
Data type	Scale
Recipes	More than 1 million
Images	More than 13 million
Ingredients	Thousands of ingredient concepts

- Supports ingredient recognition from food images.
- Supports linking a food image to a recipe or ingredient context.
- Can enrich recognition models with semantic information that may not be obvious from visual appearance alone.
- Does not directly provide the same measured portion ground truth as Nutrition5k.

For VitaMate, Recipe1M+ can strengthen food and ingredient recognition, while Nutrition5k is more directly useful for mass and nutrition estimation. The two types of datasets are complementary rather than interchangeable.

7.6 NutritionEstimation / Nutrition5k-Based Nutrient Estimation

The original study describes NutritionEstimation as a deep-learning approach that analyzes meal images, identifies foods, estimates quantities, and then calculates nutritional values. The described implementation uses computer vision and deep networks such as ResNet/CNN backbones, food segmentation, and volume or mass estimation, with Nutrition5k as a principal training source.

Element	Description
Model name	NutritionEstimation
Field	Computer Vision for Dietary Assessment
System type	AI system for meal-image analysis
Learning type	Deep Learning
Framework	PyTorch (as stated in the original study)
Goal	Estimate meal nutritional values from images

Training Data and Outputs

- Food images and ingredient weights.
- Calories, protein, fat, carbohydrates, sodium, and fiber.
- Estimated total food weight / mass.
- Nutrition5k-based supervision for meal composition and ground-truth quantities.

Performance Values Preserved from the Original Study

The following values are retained from the original reference study. They correspond to Nutrition5k-related nutrient-estimation baselines and should be interpreted in the context of the specific dataset and evaluation protocol rather than as universal accuracy guarantees.

Metric	Reported value
Calorie MAE	≈ 47.6 kcal
Food-mass MAE	≈ 29.4 g
Protein mean error	≈ 3.7 g
Fat mean error	≈ 2.27 g
Carbohydrate mean error	≈ 4.6 g

Recent related research has continued to improve this class of methods. For example, depth-prediction and fusion approaches have reported lower errors on Nutrition5k under their own experimental settings, reinforcing the value of geometry/depth cues for portion estimation. However, the practical mobile workflow must balance accuracy against capture complexity and device requirements.

For VitaMate, the most important design implication is not to present estimated calories or macronutrients as exact measurements. The system should expose uncertainty, allow correction, and use the confirmed result for final logging.

7.7 Food Detection Using YOLOv5

The original reference study includes a YOLOv5-based food-detection example as a fast object-detection layer for locating and classifying foods inside meal images. YOLO-style detectors are relevant when a meal contains several visually separable components and the system needs real-time detection before nutritional mapping.

- Input: a meal image.
- Image is processed by a YOLOv5 detector.
- Food objects are localized with bounding boxes.
- Each detected component is assigned a food class.
- Detected food identities can then be passed to a nutrition database or a second model for portion/nutrient estimation.

Datasets Mentioned in the Original Study

- Food-101.
- UECFOOD-256.
- Custom food-image datasets with labeled food classes.

Performance Values Preserved from the Original Study

Metric	Value
Detection Accuracy	85%–92%
mAP (mean Average Precision)	0.88
Precision	0.90
Recall	0.87
Processing speed	30–45 FPS

Strengths and Limitations

- Fast food detection and potential real-time execution.

- Suitable as one stage in a mobile pipeline.
- Easy to connect to a downstream nutrition-lookup component.
- Does not independently estimate food quantity accurately.
- Performance can decrease for visually similar foods and mixed dishes.
- Requires a second estimation or review step before reliable nutrition logging.

7.8 Additional Recent Research Directions

The original AI food-analysis section was strengthened with additional recent references that support the same technical direction without introducing an AI diet-recommendation service.

- DPF-Nutrition (2023) combines depth prediction and multi-scale fusion for nutritional estimation and demonstrates that learned depth cues can improve calorie, mass, and macronutrient prediction on Nutrition5k.
- goFOOD (2023) demonstrates a smartphone-oriented pipeline that combines food segmentation, recognition, volume estimation, and nutritional calculation under free-living conditions, illustrating the importance of end-to-end evaluation outside laboratory-only datasets.
- SNAPMe (2023) provides a benchmark dataset of food photos for dietary assessment and highlights the continuing need for diverse, well-annotated image datasets.
- Recent evaluations of commercial AI-enabled food-recognition applications show that image-based nutritional outputs can vary substantially across apps and meal types, supporting VitaMate's decision to require user review before finalization.

These studies reinforce a consistent conclusion: AI can substantially reduce food-logging effort, but nutritional estimation from a single image remains uncertain. The safest engineering pattern is therefore assistive analysis with explicit review and correction rather than silent, fully autonomous logging.

7.9 Comparative Summary of AI Food-Analysis References

Study / System	Model / Method	Inputs	Task / Output	Performance / Notes	Value to VitaMate
Nutrition5k	Reference dataset; classification / segmentation / volume / nutrient estimation	RGB, RGB-D, ingredient weights, metadata	Training and evaluation for calories/macros/components	Strength is measured ingredient-weight ground truth; dataset scale and annotation noise remain limitations	Strong foundation for training/evaluation
2DNCP	ResNet-50/101, InceptionV3, ViT-B/16 + regression	2D meal images linked to Nutrition5k targets	Mass and nutritional composition prediction	ResNet-101 and ViT-B/16 were among strongest in referenced comparison	Useful for backbone selection
FRCES	EfficientNet + YOLOv8 + nutrition lookup	Meal image	Food type, components, portion, calories/macros	No single verified end-to-end performance value in original study	Architecturally similar to a staged VitaMate service
Im2Calories	CNN + segmentation + volume estimation + database	Food image	Food recognition, portion, calories	Foundational system; original study did not include detailed benchmark table	Useful historical reference for image→portion→calorie pipeline
Recipe1M+	Large cross-modal food/recipe dataset	>1M recipes, >13M images, ingredient text	Ingredient and recipe understanding	Not a direct portion-estimation model	Useful for recognition/context enrichment
NutritionEstimation	ResNet/CNN + segmentation + volume estimation	Food image; Nutrition5k supervision	Calories, macros, mass	Original study: ≈ 47.6 kcal MAE, ≈ 29.4 g mass MAE; macro errors also reported	Strong practical reference for nutrient estimation
YOLOv5 food detection	Real-time food object detection	Meal image; Food-101/UECFOOD/custom datasets	Food localization and classification	Original study reports 85–92% detection accuracy, mAP 0.88, precision 0.90, recall 0.87, 30–45 FPS	Useful as fast detection stage; requires separate quantity estimation

7.10 Capability Matrix

Capability	N5K	2DNCP	FRCES	I2C	R1M+	NE	FD-Y5
Food image as primary input	✓	✓	✓	✓	✓	✓	✓
Food / ingredient recognition	✓	△	✓	✓	✓	✓	✓
Portion size / weight estimation	✓	✓	✓	✓	✗	✓	✗
Calorie estimation	✓	✓	✓	✓	△	✓	△
Macronutrient estimation	✓	✓	✓	△	△	✓	△
Reference dataset / nutrition database linkage	✓	✓	✓	✓	✓	✓	✓
Real-time suitability	✗	✗	△	✗	✗	✗	✓
Direct usefulness in VitaMate pipeline	✓	✓	✓	✓	△	✓	✓

Legend: ✓ supported / central △ partial or indirect ✗ not a central capability

8. Synthesis and Relevance to VitaMate

The reference applications confirm that strong specialized solutions exist for individual health domains. Nutrition applications provide detailed nutrient tracking and increasingly support image-based logging; medication applications offer mature reminder and adherence workflows; hydration and sleep applications optimize specialized daily interactions; and activity applications demonstrate progress visualization, challenges, and social motivation. The main opportunity for VitaMate is not to outperform each specialized application in every individual feature, but to coordinate a broader set of health behaviors within one coherent platform.

The updated motivation review shows that points, streaks, missions, badges, progress indicators, and reminders can support engagement when they reinforce meaningful behavior. Duolingo, Habitica, and Finch provide complementary cross-domain benchmarks: Duolingo demonstrates high-frequency progress loops, Habitica demonstrates explicit task-to-reward progression and the risks of misaligned incentives, and Finch demonstrates gentle self-care reinforcement and flexible recovery from missed routines. VitaMate therefore uses motivation as a support layer around verified health actions, not as a competitive game or a replacement for health-state evaluation.

The AI food-analysis literature shows that reliable nutrition estimation requires more than simple image classification. Recognition, segmentation, portion/mass estimation, nutrition-database mapping, and uncertainty management must work together. The most appropriate design for VitaMate is therefore a review-assisted workflow in which AI provides provisional information and the user confirms the final record.

The former AI-generated diet-recommendation section was intentionally removed from this edition. Chronic-condition handling remains important in VitaMate, but it is represented through explicit constraints and system logic rather than an unverified AI service that generates therapeutic meal plans. This distinction improves technical accuracy, safety, and alignment between the reference study and the implemented project scope.

References

- [1] World Health Organization, WHO Clinical Treatment Guideline for Tobacco Cessation in Adults. Geneva, Switzerland: World Health Organization, 2024.
- [2] NHS Better Health, “Managing nicotine withdrawal symptoms,” National Health Service.
- [3] NHS, “Foods to avoid in pregnancy,” National Health Service.
- [4] R. Whittaker et al., “Mobile phone text messaging and app-based interventions for smoking cessation,” Cochrane Database of Systematic Reviews, no. 10, CD006611, 2019. doi: 10.1002/14651858.CD006611.pub5.
- [5] L. F. Stead et al., “Combined pharmacotherapy and behavioural interventions for smoking cessation,” Cochrane Database of Systematic Reviews, no. 3, CD008286, 2016. doi: 10.1002/14651858.CD008286.pub3.
- [6] EFSA Panel on Dietetic Products, Nutrition and Allergies, “Scientific Opinion on the safety of caffeine,” EFSA Journal, vol. 13, no. 5, 4102, 2015. doi: 10.2903/j.efsa.2015.4102.
- [7] Food Standards Agency, “Food supplements containing caffeine,” 2024.
- [8] E. T. Hébert et al., “Just-in-Time Adaptive Intervention for Smoking Cessation in Low-Income Adults: A Randomized Clinical Trial,” JAMA Network Open, vol. 8, no. 8, e2526691, 2025. doi: 10.1001/jamanetworkopen.2025.26691.

- [9] J. B. Bricker et al., “Efficacy of Smartphone Applications for Smoking Cessation: A Randomized Clinical Trial,” *JAMA Internal Medicine*, vol. 180, no. 11, pp. 1472–1480, 2020. doi: 10.1001/jamainternmed.2020.4055.
- [10] C. Free et al., “Smoking cessation support delivered via mobile phone text messaging (txt2stop): A single-blind, randomised trial,” *The Lancet*, vol. 378, no. 9785, pp. 49–55, 2011. doi: 10.1016/S0140-6736(11)60701-0.
- [11] S. Glasner et al., “Long-Term Effectiveness of a Clinician-Assisted Digital Cognitive Behavioral Therapy Intervention for Smoking Cessation,” *Nicotine & Tobacco Research*, vol. 24, no. 11, pp. 1763–1772, 2022. doi: 10.1093/ntr/ntac113.
- [12] E. A. Edwards et al., “Gamification for health promotion: systematic review of behaviour change techniques in smartphone apps,” *BMJ Open*, vol. 6, no. 10, e012447, 2016. doi: 10.1136/bmjopen-2016-012447.
- [13] L. Sardi, A. Idri, and J. L. Fernández-Alemán, “A systematic review of gamification in e-Health,” *Journal of Biomedical Informatics*, vol. 71, pp. 31–48, 2017. doi: 10.1016/j.jbi.2017.05.011.
- [14] L. Xu et al., “The Effects of mHealth-Based Gamification Interventions on Participation in Physical Activity: Systematic Review,” *JMIR mHealth and uHealth*, vol. 10, no. 2, e27794, 2022. doi: 10.2196/27794.
- [15] S. Tran, L. Smith, S. El-Den, and S. Carter, “The Use of Gamification and Incentives in Mobile Health Apps to Improve Medication Adherence: Scoping Review,” *JMIR mHealth and uHealth*, vol. 10, no. 2, e30671, 2022. doi: 10.2196/30671.
- [16] S. K. Nishi et al., “Effect of digital health applications with or without gamification on physical activity and cardiometabolic risk factors: a systematic review and meta-analysis of randomized controlled trials,” *eClinicalMedicine*, vol. 76, 102798, 2024. doi: 10.1016/j.eclinm.2024.102798.
- [17] R. Hadi Mogavi et al., “When Gamification Spoils Your Learning: A Qualitative Case Study of Gamification Misuse in a Language-Learning App,” in *Proc. 9th ACM Conference on Learning @ Scale*, 2022, pp. 175–188. doi: 10.1145/3491140.3528274.
- [18] Duolingo, “The Duolingo Streak Uses Habit Research to Keep You Motivated,” *Duolingo Blog*, 2021.
- [19] Duolingo, “How to Use Duolingo for Language Learning,” *Duolingo Blog*, updated product guide.
- [20] Duolingo, “Understanding Duolingo’s Time Spent Learning Well Metric,” *Duolingo Blog*.
- [21] Q. Thames et al., “Nutrition5k: Towards Automatic Nutritional Understanding of Generic Food,” in *Proc. IEEE/CVF CVPR*, 2021, pp. 8903–8911.
- [22] A. Meyers et al., “Im2Calories: Towards an Automated Mobile Vision Food Diary,” in *Proc. IEEE ICCV*, 2015, pp. 1233–1241.
- [23] A. Salvador et al., “Learning Cross-Modal Embeddings for Cooking Recipes and Food Images,” in *Proc. IEEE CVPR*, 2017, pp. 3020–3028. doi: 10.1109/CVPR.2017.327.
- [24] R. Bianco et al., “2D Prediction of the Nutritional Composition of Dishes from Food Images: Deep Learning Algorithm Selection and Data Curation Beyond the Nutrition5k Project,” *Nutrients*, vol. 17, no. 13, 2196, 2025. doi: 10.3390/nu17132196.
- [25] U.S. Department of Agriculture, Agricultural Research Service, “FoodData Central.”
- [26] L. Wang et al., “DPF-Nutrition: Food Nutrition Estimation via Depth Prediction and Fusion,” *Foods*, vol. 12, no. 23, 4293, 2023. doi: 10.3390/foods12234293.
- [27] goFOOD research team, “The Nutritional Content of Meal Images in Free-Living Conditions—Automatic Assessment with goFOOD,” *Nutrients*, vol. 15, no. 17, 3835, 2023. doi: 10.3390/nu15173835.

- [28] “Surveying Nutrient Assessment with Photographs of Meals (SNAPMe): A Benchmark Dataset of Food Photos for Dietary Assessment,” *Nutrients*, vol. 15, no. 23, 4972, 2023. doi: 10.3390/nu15234972.
- [29] MyNetDiary.com, “Calorie Counter – MyNetDiary” and “Plate AI – Calorie Tracker,” official/store listings.
- [30] Cronometer Software Inc., “Calorie Counter by Cronometer,” official/store listing.
- [31] Nutrivine, “Nutrivine – Smart Nutrition Tracking,” official website and App Store listing.
- [32] Microgram, “Microgram – Advanced Nutrient Tracking App,” official website and App Store listing.
- [33] Foodvisor, “Foodvisor – AI Calorie Counter,” official/store listing.
- [34] Medisafe, “Medisafe Pill & Med Reminder,” official/store listing.
- [35] MyTherapy, “Meds & Pill Reminder MyTherapy,” official/store listing.
- [36] Funn Media, “Water Tracker: WaterMinder,” official/store listing.
- [37] Codium App Ideas, “Water Tracker – Hydro Coach,” official/store listing.
- [38] Sleep Cycle AB, “Sleep Cycle: Sleep Tracker,” official/store listing.
- [39] Neybox Digital Ltd., “Pillow: Sleep Tracker,” App Store listing.
- [40] Strava Inc., “Strava: Run, Bike, Hike,” official/store listing.
- [41] Google LLC, “Google Fit: Activity Tracking,” store listing referenced by the original study.
- [42] despDev, “Quitzilla,” Google Play listing.
- [43] Kanza Ahmad, “Quit Smoking Tracker – BeSober,” App Store listing.
- [44] DevLab Spa, “We Quit!,” App Store listing.
- [45] “Addicted: Quit Habit Tracker,” App Store listing.
- [46] QuitWithUs, “QuitWithUs – Quit Habits Together with Community Support,” official website and App Store listing.
- [47] Duolingo, official product and brand documentation, including Duolingo 101, streak, quest, and leaderboard documentation.
- [48] Habitica, “Features: Gamify Your Life,” Habitica, official product documentation, accessed Aug. 2026.
- [49] C. Gerdenitsch, D. Sellitsch, M. Besser, S. Burger, C. Stegmann, M. Tscheligi, and S. Kriglstein, “Work gamification: Effects on enjoyment, productivity and the role of leadership,” *Electronic Commerce Research and Applications*, vol. 43, 100994, 2020. doi: 10.1016/j.elerap.2020.100994.
- [50] S. Diefenbach and A. Müssig, “Counterproductive effects of gamification: An analysis on the example of the gamified task manager Habitica,” *International Journal of Human-Computer Studies*, vol. 127, pp. 190–210, 2019. doi: 10.1016/j.ijhcs.2018.09.004.
- [51] Finch, “Our Approach to Self-Care,” Finch Help Center, updated July 2025, accessed Aug. 2026.
- [52] Finch, “Daily Quests,” Finch Help Center, updated July 2026, accessed Aug. 2026.
- [53] Finch, “Understanding Streaks,” Finch Help Center, updated July 2025, accessed Aug. 2026.